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DATABASE System

SHORT QUESTION...!

QUESTION: 1
Explain Transaction and Acid Properties of Transaction:

A transaction is defined as a logical unit of work. It consists of one or more operations on a database that must be completed together so that the database remains in a consistent state. A transaction must either complete successfully or fail.

PROPERTIES

— ATOMICITY:

A transaction must be atomic unit of work. It must completely succeed or completely fail. If any statements of transaction fail, the entire transaction fails completely.

When transaction is executed successfully, it's said to be committed. In case of failure of a statement, all previous successful statements in transaction are rolled back.

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CONSISTENCY:

A transaction most leave the data in a consistent state after completion. e.g in a bank database money should never be "created" or "deleted" without an appropriate deposit or withdrawal.

ISOLATION:

All transaction that modify the data are isolated from each other. They don't access the same data at the same time. Transaction most have no dependence or effect on another transaction.

DURABILITY:

The durability means that the modification made by a transaction are permanent and persistent. If the system is crashed or rebooted, data should be guarnteed to be completed when the computer restart.

Question: 2

Related consistency, Redudancy and normalization?

NORMALIZATION

The Process of Producing a simpler and more reliable database structure is called normalization. It is used to create a suitable set

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of relations for storing data. This Process works through different stages known as normal form. These stages are 1NF, 2NF, 3NF and so on.

REDUNDANCY

Redundancy mean control on the same data. It exists when the same data is stored unnecessarily at different Places

RELATED CONSISTENCY

It refers to maintaining the accuracy and integrity of data across different table that are connected through relationship.

when database have relationship between table such as foreign key linking record in one table to record in another, related consistency ensures these relationships are maintained correctly.

Que: 3

Explain Relational Database?

A relational database is a type of database that organizes data into rows and columns, which collectively form a table where the data points are related to eachother

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Data is typically structured across multiple tables, which can joined together via a primary key or a foreign key.

Que: 4 Explain candidate key and its two properties:

A candidate key is a unique identifier for a record within a table that can be chosen as a primary key.

It has two properties :-

→ uniqueness

→ Minimality

Uniqueness

Each value or combination of value in candidate key must be uniqueness across all row in table

Minimality

The candidate key should be minimal, meaning that no subset of candidate key should have uniqueness property

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Question: 5 Explain ANSI sprach Architecture layer?

ANSI stand for "American national standard institute". It has three layers:

→ External schema
→ conceptual schema
→ Internal schema

External schema:

It's a user view representation, how user view the database. An external schema ~~property~~ portrays just a portion of database.

conceptual schema:

It is a complete logical view of database that contain description of all data and relationships in a database. It is independent of any particular means of storing data.

Internal schema:

It is a representation of a conceptual schema as physically stored using a particular product and/or technique. The description of a set of tables, keys, foreign keys, indexes and other physical structures is an internal schema

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Q:2

A sale person may manage many other salespeople. A sales person is managed by only one salespeople. A salesperson can be agent for many customer. A customer is managed by one sales ~~person~~ people. A customer can placed many order. An order can be placed one customer. An order lists many inventory item. An inventory items may be listed on many orders. An inventory item is assembled from many parts. A part may be assembled into many inventory items. Many employees assembles an inventory item from many parts. A parts may be supplied by many suppliers

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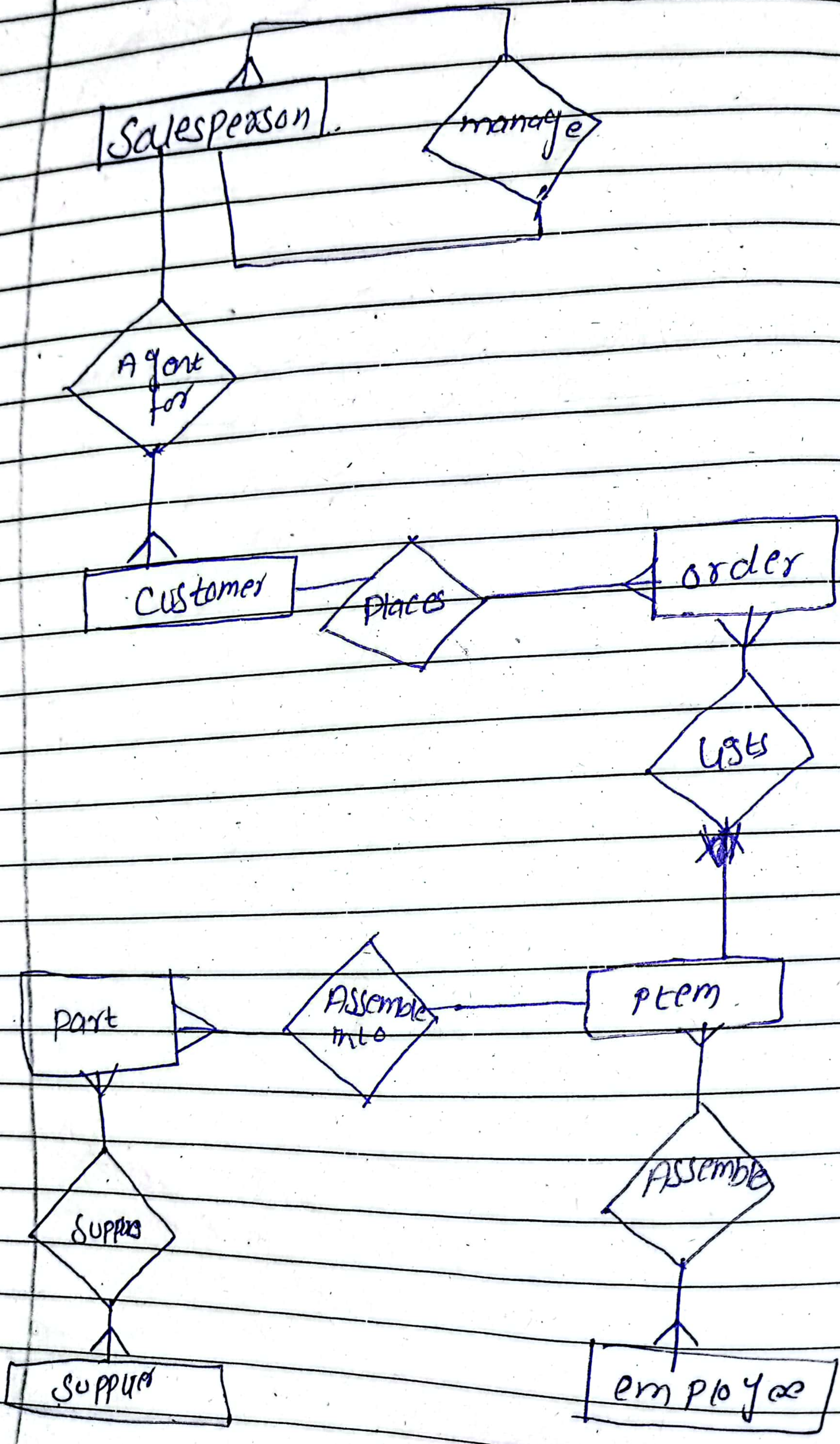

CamScanner

Q:3
Explain NORMALIZATION form

The process of producing a simpler and more reliable database structure is called normalization. It is used to create a suitable set of relations for storing data. This process work through different stages known as normal form. stages are 1NF, 2NF, 3NF and so on.

FIRST NORMAL FORM

→ It doesn't contain repeating group. A repeating group is a set of one or more data item that may occur a variable number of item in a tuple.

SECOND NORMAL FORM

A relation in a 2NF if it is in 1NF and if all of its non-key attributes are fully functionally dependent on the whole key. It means the none of the non-key attributes are related to a part of key

THIRD NORMAL

A relation is in third normal form if it is 2NF and if non-key attributes is dependent on another non-key attribute. It means that all non-key attributes are

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functionally dependent only on primary key.

Boyce-codd Normal Form

A relation is a BCNF if and only if every determinant is a candidate key. It can be checked by identifying all determinants and then making sure that all the determinants are candidate key.

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Q:4 write SQL statement

Title	Sal
Elect . Eng	40,000
Syst . Anal	34,000
Mech. Eng	27,000
Programmer	24,000

1)

ENO	Ename	Title	Hire Date
E1	John	Elec. Eng	22/11/1990
E2	Smith	Syst. Anal	27/6/1990
E3	Lee	Mech. Eng	15/12/1990
E4	Michel	Programmer	30/5/1990
E5	Miller	Syst. Anal	11/2/1991
E6	Davis	Elec. Eng	7/2/1993
E7	Jones	Mech. Eng	17/5/1994
E8	Cathey	Syst. Anal	27/2/1991

Employee

Eno	Pno	Resp	Dur
E1	P1	Manager	12
E2	P1	Analyst	24
E2	P2	Analyst	8
E3	P3	Consultant	24
E3	P4	Engineer	16
E4	P2	Programmer	18
E8	P2	Manager	0
E6	P4	Engineer	36
E7	P1	Engineer	0

ASG

PROJ

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Pno	Pname	Budget
P1	Instrumentation	150000
P2	Database prog...	135000
P3	CAD/CAM	250000
P4	Maintenance	310000

1) Show salary of programmer working on instrumentation project.

```
Select SAl
From Employee E
Join ASG A ON Eno = A.Eno
Join Proj P ON A.Pno = P.Pno
where E.Title = "programmer" AND
P.Pname = "Instrumentation";
```

11) Show All record of ASG table, which is not appear in equi-join with EMP, PROJ.

```
Select *
From ASG A
WHERE NOT EXISTS (
Select 1
From Employee E
Join PROJ P ON A.Pno = P.Pno
where E.Eno = A.Eno
```

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3) SHOW The name of Hire date is a format (24th Dec 1994) whose salary is greater than the salary of Manager But less than that of Consultant and not Equal to that programmer.

```
Select Ename, TO_CHAR (Hiredate, DDth, Month YYYY) As Hire date
From Employee
where Sal > (
   Select Sal
    From Title
    WHERE Title = 'Manager'
)
```

```
AND Sal < (
 select sal
  From Title
  where title = "consultant"
)
 And Sal != (
  Select Sal
  FROM Title
  where Title = 'programmer'
);
```

4) Show the total salary (i.e sal + Bonus) of all the employee, values display in a format (40000 + 3000 = 43000) with the heading Total salary

```
Select CONCAT(sal,'+', Bonus, '=',
(Sal + Bonus)) As "Total salary"
From Employee;
```

Note : Assuming the Bonus column exists, if not, you might need to consider a default bonus or a calculated Bonus.

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5. Show name of those hired before 1995 and their name having A and F and Display substring of name from A to F.

```
Select (SUBSTR( Ename, INSTR(Ename,'A'),
INSTR( Ename,'F') - INSTR(Ename,'A')+1)
As subname
From Employee;
   Where Hiredate <
~~Note~~: To-Date('01-jan-1995',
 DD-MON-YYYY)
And Ename like '%A%F';
```